Simple Linear Regression

Chapter 3.1 — An Introduction to Statistical Learning

A foundational approach for predicting a quantitative response Y using a single predictor variable X via a linear relationship.

Topics: Model Setup · Coefficient Estimation · Standard Errors · Hypothesis Testing · R^2 & RSE

The Linear Model

$$Y \approx \beta_0 + \beta_1 X$$

$$Y_i = \beta_0 + \beta_1 \cdot X_i + \varepsilon_i$$

Model Components

β_0	Intercept — expected value of Y when X = 0
β_1	Slope — avg. increase in Y per unit increase in X
ε	Error term — captures unexplained variation
$\hat{Y}$	Predicted value: $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x$

$\rightarrow \mathbb{E}[\varepsilon_i] = 0, \ \mathrm{Var}[\varepsilon_i] = \sigma^2$

$\varepsilon_i \overset{i.i.d}{\sim} N(0, \sigma^2)$ and

$Y_i \overset{\text{indep}}{\sim} N(x_i\beta_1 + \beta_0, \sigma^2)$, $\begin{cases} \mathbb{E}[Y_i] = \beta_0 + \beta_1 x_i \\ \mathrm{Var}[Y_i] = \sigma^2 \end{cases}$

Advertising Example

sales $\approx \beta_0 + \beta_1 \times$ ***TV***

- X = TV advertising budget ($1,000s)
- Y = Product sales (1,000 units)

Goal: Find β_0 and β_1 such that the line fits the data as closely as possible.

Estimating the Coefficients

Residual: $e_i = y_i - \hat{y}_i$ (difference between observed and predicted value for observation i)

$= y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)$

Residual Sum of Squares (RSS)

$$RSS = e_1^2 + e_2^2 + \cdots + e_n^2$$

Least Squares Approach

1. Compute $e_i = y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)$ for each of the n observations
2. Square each residual and sum them to get RSS
3. Choose $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimize RSS — this is the least squares criterion

↳ how...?

Least Squares Coefficient Estimates

Slope $\hat{\beta}_1$

$$\hat{\beta}_1 = \frac{\Sigma(x_i - \bar{x})(y_i - \bar{y})}{\Sigma(x_i - \bar{x})^2}$$

Intercept $\hat{\beta}_0$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Key Intuitions

- $\bar{x}$ and $\bar{y}$ are sample means: $\bar{x} = (1/n)\Sigma x_i$ and $\bar{y} = (1/n)\Sigma y_i$
- β_1 measures how much Y co-varies with X, normalized by the variance of X
- β_0 ensures the regression line always passes through the point $(\bar{x}, \bar{y})$

Past exam 39th Q4 (1)

Advertising Example: Fitted Model

$$sales \approx 7.03 + 0.0475 \times TV$$

Intercept

$$\hat{\beta}_0 = 7.03$$

Baseline sales when TV advertising budget = $0

Slope

$$\hat{\beta}_1 = 0.0475$$

Each additional $1,000 in TV advertising → ~47.5 more units

Interpretation: For every $1,000 increase in TV advertising, product sales increase by approximately 47.5 units on average. Even with zero advertising, the model predicts ~7,030 baseline sales.

Population Line vs. Least Squares Line

Population Regression Line

- True (unknown) linear relationship: $Y = \beta_0 + \beta_1 X + \varepsilon$
- β_0 and β_1 are fixed but unknown constants
- Cannot be observed in practice
- Represents the best linear approximation to f(X)

In general,

$\beta_0 \neq \hat{\beta}_0$

$\beta_1 \neq \hat{\beta}_1$

Least Squares Line

- Our best estimate using observed data
- $\hat{\beta}_0$ and $\hat{\beta}_1$ are computed from training data
- Changes slightly with each sample
- On average, very close to the true population line

Analogy: Just as we estimate the population mean μ with the sample mean $\hat{\mu}$, we estimate β_0 and β_1 with β_0 and β_1. Both estimates are unbiased — they don't systematically over- or under-estimate the true value.

Standard Errors of Coefficient Estimates

$SE(\hat{\beta}_1)^2$

$$\sigma^2 \;/\; \Sigma(x_i - \bar{x})^2$$

Smaller when X values are more spread out

$SE(\hat{\beta}_0)^2$

$$\sigma^2 \cdot [1/n + \bar{x}^2 / \Sigma(x_i - \bar{x})^2]$$

Equals $SE(\hat{\mu})^2$ when $\bar{x} = 0$

where $\sigma^2 = Var(\varepsilon)$ and σ is estimated by $RSE = \sqrt{(RSS / (n-2))}$

95% Confidence Intervals

$$\hat{\beta}_1 \pm 2 \cdot SE(\hat{\beta}_1) \qquad \hat{\beta}_0 \pm 2 \cdot SE(\hat{\beta}_0)$$

Advertising data: $\beta_1 \in [0.042, 0.053]$ | $\beta_0 \in [6.130, 7.935]$

why 2...?

Hypothesis Testing

Ho (Null) $\beta_1 = 0 \rightarrow$ *No relationship between X and Y*

H_a (Alt.) $\beta_1 \neq 0 \rightarrow$ *Some relationship exists*

$$Z \sim N(0,1)$$
$$V \sim \chi^2(v) \triangleq Z_1^2 + Z_2^2 + \cdots + Z_v^2$$
$$t := \frac{Z}{\sqrt{V/v}} \sim t(v)$$

t-Statistic

$t = \hat{\beta}_1 - 0 \;/\; SE(\hat{\beta}_1) \rightarrow$ *measures how many standard deviations $\hat{\beta}_1$ is from 0*

Advertising Data Results

Term	Estimate	Std. Error	t-Statistic	p-value
Intercept	7.0325	0.4578	15.36	< 0.0001
TV	0.0475	0.0027	17.67	< 0.0001

Assessing Model Fit: RSE and R^2

Residual Standard Error (RSE)

$$RSE = \sqrt{(RSS / (n-2))}$$

- Estimates σ (std. dev. of error ε)
- Average deviation of Y from regression line
- Advertising data: RSE = 3.26 (~23% of mean sales)

R^2 Statistic

$$R^2 = (TSS - RSS) / TSS$$

- Proportion of variance explained (0 to 1)
- $R^2 = 1 - RSS/TSS$ where $TSS = \Sigma(y_i - \bar{y})^2$
- Advertising data: $R^2 = 0.61$ (61% explained)



RSE vs. R^2 — Key Difference

RSE: Absolute measure (in units of Y) — context-dependent. **R^2:** Relative proportion — always between 0 and 1, scale-independent. In simple regression, $R^2 = r^2$ (squared correlation).

Chapter 3.1 — Key Takeaways

Model	SLR models Y as a linear function of X: $Y \approx \beta_0 + \beta_1 X$
Estimation	Coefficients β_0 and β_1 are found by minimizing RSS (least squares)
Accuracy	Standard errors (SE) measure precision of coefficient estimates
Testing	t-statistics and p-values test whether X truly relates to Y
Fit	RSE (absolute) and R^2 (proportional) measure overall model quality